

LIPERI TUISKAVANLUOTO, Finland

WMO: 027930

Lat: 62.546N

Lon: 29.668E

Elev: 261

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-10.4	-21.6	1.7	-16.4	-14.9	2.5	-10.3	29.5	22.0	27.6	24.1	2.6	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	78.4	68.5	75.5	66.7	72.5	65.3	70.4	75.9	68.3	73.8	66.1	71.8	8.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	105.1	74.4	66.1	97.0	72.3	63.5	88.7	69.8	34.5	75.8	32.7	74.0	30.9	71.7	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	24.8	22.5	DB	-18.3	80.2	9.8	4.5	-25.3	83.5	-31.1	86.1	-36.6	88.7	-43.7	91.9	(4)
(5)			WB	-19.0	70.6	9.2	4.5	-25.6	73.8	-30.9	76.5	-36.1	79.0	-42.7	82.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	15.6	18.7	23.5	34.0	48.6	57.9	64.9	61.7	53.1	40.8	32.4	24.8	
	DBStd	18.68	13.42	12.87	9.52	5.95	8.17	6.78	6.34	5.59	5.66	5.80	6.81	11.23	
	HDD50	5011	1067	876	820	479	126	11	0	1	36	286	528	781	
	HDD65	9328	1532	1296	1285	929	509	230	82	133	359	749	978	1246	
	CDD50	1288	0	0	0	0	84	249	462	363	128	2	0	0	
	CDD65	127	0	0	0	0	1	17	79	30	0	0	0	0	
	CDH74	417	0	0	0	0	2	59	288	68	0	0	0	0	
	CDH80	43	0	0	0	0	0	8	32	3	0	0	0	0	
Wind	WSAvg	11.7	11.3	11.3	11.7	9.9	10.4	11.3	10.3	11.0	12.5	13.8	13.6	12.8	
Precipitation	PrecAvg	24.4	1.9	1.5	1.3	1.2	1.6	2.4	2.7	2.9	2.0	2.5	2.2	2.0	
	PrecMax	28.7	3.1	2.6	2.8	2.6	3.0	4.8	5.3	7.4	5.5	5.0	5.1	3.2	
	PrecMin	20.9	0.6	0.2	0.3	0.2	0.5	1.3	1.1	1.1	0.8	0.5	0.6	0.8	
	PrecStd	2.7	0.7	0.7	0.5	0.6	0.7	1.0	1.1	1.6	1.1	1.2	1.0	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	38.6	41.4	52.3	72.6	80.0	83.6	79.1	67.2	53.8	47.3	40.8	
		MCWB	35.3	36.0	36.3	43.8	62.1	66.9	70.5	69.4	61.2	51.8	45.5	39.9	
	2%	DB	34.9	35.7	38.4	47.1	68.3	74.8	80.2	75.8	63.8	51.8	44.0	38.0	
		MCWB	33.8	34.4	34.7	40.7	58.4	64.7	69.4	67.1	59.6	49.9	42.7	36.7	
	5%	DB	33.6	34.2	36.4	43.7	63.8	70.6	77.1	72.5	62.0	50.2	42.4	36.2	
		MCWB	33.0	33.4	33.9	38.8	56.2	63.3	68.4	65.3	58.5	48.2	41.6	35.2	
	10%	DB	32.3	33.2	34.9	41.1	60.2	67.3	74.5	69.6	60.4	48.5	40.1	34.8	
		MCWB	31.8	32.5	33.2	37.0	53.3	60.3	66.9	63.7	57.1	46.5	39.3	34.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	36.2	37.1	44.9	62.2	69.9	73.1	72.3	63.3	52.9	45.4	39.9
			MCDB	36.5	37.4	39.2	50.2	69.3	75.5	79.7	77.3	65.1	53.5	46.6	40.5
2%		WB	34.1	34.7	35.6	41.7	59.8	66.4	71.3	68.2	61.1	50.7	43.0	37.0	
		MCDB	34.7	35.5	37.2	46.2	65.9	72.8	77.1	73.3	62.9	51.5	43.8	37.6	
5%		WB	33.1	33.5	34.4	39.4	57.0	63.7	69.7	66.1	59.2	48.4	41.6	35.3	
		MCDB	33.6	34.1	35.8	42.9	63.4	69.7	75.0	71.4	61.2	49.6	42.5	35.9	
10%		WB	31.9	32.6	33.4	37.7	54.0	60.9	68.0	64.3	57.6	46.7	39.4	34.1	
		MCDB	32.5	33.1	34.7	40.3	59.5	66.4	73.7	68.9	59.7	48.3	39.9	34.6	
Mean Daily Temperature Range		5% DB	MDBR	10.8	11.0	13.7	11.8	11.3	9.9	9.7	8.1	6.6	5.4	4.9	6.7
			MCDBR	5.7	7.9	11.8	14.9	16.3	14.1	12.8	10.9	8.4	6.1	6.1	4.8
	MCWBR		5.5	7.5	9.7	9.3	8.6	8.0	6.9	6.1	5.6	5.5	5.7	4.8	
	5% WB	MCDBR	5.5	7.2	9.7	12.8	15.3	13.3	11.8	9.5	7.3	6.1	6.0	4.8	
		MCWBR	5.4	6.7	8.5	8.5	9.0	8.2	7.3	5.9	5.8	6.0	5.7	4.8	
Clear-Sky Solar Irradiance	taub	0.235	0.288	0.315	0.354	0.353	0.348	0.374	0.357	0.335	0.314	0.252	0.214		
	taud	2.009	2.107	2.178	2.315	2.446	2.489	2.431	2.462	2.511	2.397	2.103	1.973		
	Ebn at Noon	129	204	247	263	275	278	266	261	246	201	122	75		
	Edh at Noon	11	22	30	32	31	30	31	28	22	17	10	6		
All-Sky Solar Radiation	RadAvg	65	244	726	1164	1533	1685	1604	1223	679	261	67	30		
	RadStd	9	59	98	140	174	94	151	142	80	41	11	4		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	-344	N/A	N/A	N/A

Nomenclature: See separate page